

MongoDB Atlas vs Aurora PostgreSQL vs RDS PostgreSQL

Features Comparison — database capabilities, scaling, availability, security and AWS integration

1. Quick Overview

Area	MongoDB Atlas	Aurora PostgreSQL	RDS PostgreSQL
Model	Document / NoSQL	Relational / PostgreSQL-compatible	Relational / PostgreSQL
Service	Fully managed multi-cloud database service	Fully managed AWS relational database engine	Fully managed AWS PostgreSQL service
Best fit	Flexible document data, MongoDB-native features, global distributed workloads	Relational workloads needing strong availability, performance and AWS-native integration	Standard PostgreSQL workloads with AWS-managed operations
Clouds	AWS, Azure, Google Cloud	AWS	AWS

2. Core Features

Capability	MongoDB Atlas	Aurora PostgreSQL	RDS PostgreSQL
Schema	Flexible document schema	Relational schema + JSONB	Relational PostgreSQL schema + JSON/JSONB
Transactions	Multi-document transactions	ACID transactions with PostgreSQL semantics	PostgreSQL ACID transactions
Joins	Document modeling and aggregation patterns	Native SQL joins and aggregations	Native SQL joins and aggregations
Document data	Native BSON document model	JSONB for flexible, queryable data	JSON/JSONB support
Horizontal scaling	Sharding distributes data/workload across shards	Read scaling and managed storage scaling; distributed SQL options for supported workloads	Read replicas and larger instances/storage; not a MongoDB-style sharded cluster
Read replicas	Replica sets / deployment-specific read scaling	Up to 15 low-latency read replicas	Up to 15 read replicas
High availability	Replica sets plus multi-region/multi-cloud options	Distributed, fault-tolerant, multi-AZ storage architecture	Multi-AZ deployments and standby/readable standby options
Global	Multi-region and multi-cloud distribution	Aurora Global Database	Cross-region read replicas / DR patterns
Backup	Managed backups and point-in-time recovery options	Automated backups, snapshots and continuous backup to S3	Automated backups, snapshots and point-in-time restore

3. Security, Networking & Operations

Capability	MongoDB Atlas	Aurora PostgreSQL	RDS PostgreSQL
Encryption at rest	Encryption supported; depends on deployment/configuration	AWS KMS-backed encryption	AWS KMS-backed encryption
Encryption in transit	TLS	SSL/TLS	SSL/TLS
Network controls	IP access lists, private endpoints / peering	VPC, Security Groups and AWS networking	VPC, Security Groups and AWS networking
Authentication	Database users and certificate-based options	PostgreSQL authentication + IAM database authentication	PostgreSQL authentication + IAM authentication options
Monitoring	Atlas monitoring, alerts and performance tools	CloudWatch / Database Insights and Aurora monitoring	CloudWatch / Database Insights and RDS monitoring
Patching	Managed by Atlas	Managed by AWS	Managed by AWS

4. Performance, Scaling & Developer Features

Capability	MongoDB Atlas	Aurora PostgreSQL	RDS PostgreSQL
Compute scaling	Cluster tier/deployment can be adjusted; Flex and dedicated options	Push-button compute scaling; Serverless v2 for supported workloads	Push-button DB instance scaling
Storage scaling	Depends on selected deployment/configuration	Distributed storage automatically grows with data	Storage autoscaling with supported configurations
Search	MongoDB Search / Atlas Search	PostgreSQL full-text search + extensions	PostgreSQL full-text search + extensions
Vector / AI	Vector Search	pgvector / embeddings; Aurora ML capabilities	pgvector / embeddings
Extensions	MongoDB ecosystem and Atlas integrations	PostGIS, pgvector and other supported PostgreSQL extensions	PostGIS, pg_cron and other supported PostgreSQL extensions
Change / events	Change Streams	Logical replication and AWS integrations	Logical replication and AWS integrations
AWS analytics	Atlas integrations/connectors	Zero-ETL and AWS analytics integrations	AWS integrations including S3/Redshift patterns and supported zero-ETL capabilities

5. When Each One Makes Sense

- MongoDB Atlas:** Best when the application is naturally document-oriented, needs flexible schemas, MongoDB-native Search/Change Streams, or multi-cloud/multi-region deployment.
- Aurora PostgreSQL:** Best when the application is relational, transaction-heavy, and AWS-native, and needs strong availability, distributed storage and scaling capabilities.
- RDS for PostgreSQL:** Best when standard PostgreSQL is the priority and you want AWS to manage provisioning, backups, patching, Multi-AZ, monitoring and related operations.

Note: Exact feature availability can depend on AWS Region, PostgreSQL version, MongoDB Atlas cluster type and configuration.